Referee Report for "wxsum: A command for processing temperature and precipitation data"

Summary

This paper presents a Stata package, "wxsum", that aggregates daily location-coded temperature and precipitation data over a user-specified duration of time (i.e "season"). The package aggregates daily data to calculate many commonly used temperature and rainfall variables:

- **For temperature:** mean, standard deviation, skewness, maximum, growing degree days with user-specified lower and upper thresholds, killing degree days with a user-specified threshold, a categorical growing degree days variable, temperature bins, and deviations and z-scores relative to a lagged rolling window of user-specified length.
- **For rainfall:** mean, standard deviation, skewness, total rainfall, mean total monthly rainfall, number of days with rainfall exceeding or below a user-specified threshold, early-, mid-, and late-season dry spells (number of days in the beginning, middle, and end of the user-specified season with rainfall below the user-specified threshold), and deviations and z-scores relative to a lagged rolling window of user-specified length.

The package requires the daily temperature/rain dataset to be in wide format, with each row representing one location, and each column representing temperature/rainfall on a specific day. Columns must be named according to the convention "varname_YYYYMMDD" where YYYYMMDD is a string that denotes the date of a daily observation (for example, "tmp_20220301" would be temperature on the 1st of March, 2022).

In the file generated by wxsum, each row is a location ID, and there are different columns, each representing one of the summary variables listed above calculated over the specified season of the year. In cases where locations are observed over multiple years, all variables are calculated throughout the specified season separately for each year.

General Comments

This paper is well-written, transparent, and the Stata package it introduces is useful for the kinds of applications the authors discuss. Based on my verification exercise, the package works correctly. That is, it calculates the summary variables listed above correctly for both temperature and rainfall data. I do have some concerns and recommendations, which I outline below.

Recommendations

A "reshape wide" Option

The package requires daily temperature and rainfall data to be shaped in wide format with a specific naming convention. As the authors already point out, the main limitation of this requirement is that it will preclude some Stata users from loading temperature/rainfall datasets that span very long periods into Stata to run wxsum. There is, however, an additional limitation: requiring a wide format with a specific naming convention increases the pre-processing burden on users who (i) need to implement transformations of the daily temperature/rainfall observations (e.g. converting Kelvin to Celsius or taking averages of temperature across sub-daily observations); (ii) have access to long-format datasets.

In light of this limitation, I recommend that the authors incorporate an option into the package that allows users to convert long-format data to the wide format and naming convention required by wxsum. This option would require the user to specify an ID variable, the weather variable (temperature/rainfall), and the date variable from which the yyyymmdd suffix would be extracted. Adding this feature to the package would benefit (i) users with access to wide-format data who want to implement transformations to the daily weather variables before proceeding with wxsum (they would simply need to reshape the data long, implement the transformations, and then reshape it wide again); (ii) users with access to long-format data who want to use wxsum.

Deviations and z-Scores Relative to a Fixed Baseline

In the current version of the package, deviations and z-scores are calculated relative to a lagged rolling window. This is useful for some applications. However, in other applications, researchers might want to calculate deviations and z-scores relative to a fixed climate baseline (e.g. 1900 to 1950). I think this option should be incorporated into the package.

Other Comments and Suggestions

- In the current version of the package, the user-specified season is invariant across locations. There are applications where the season throughout which we want to calculate temperature and rainfall statistics varies by location. E.g. planting and harvest dates, and therefore growing seasons, can vary across farms. Is it possible to allow users to specify the beginning and end dates of the season based on variables in the dataset? Using the agriculture example, if we observe planting date and harvest date for each farm, this option would allow us to calculate temperature and rainfall summaries throughout farm-specific growing seasons. If implementing this or some version of it is not feasible, the authors should explain why and discuss this as a limitation.
- I noticed that deviations and z-scores are not calculated for early-, mid-, and late-season dry spells. These should either be calculated or there should be an explanation as to why they are not.